

Annotation noise-aware semi-supervised and domain adaptive learning for 3D medical imaging

Suruchi Kumari , Pravendra Singh* 

Department of Computer Science and Engineering, Indian Institute of Technology Roorkee, Roorkee, India

HIGHLIGHTS

- A survey of recent advances in deep unsupervised domain adaptation (UDA) for medical imaging.
- A Comprehensive overview of more than 140 relevant papers is presented.
- The UDA techniques are categorized into six groups and divided into finer subcategories based on their different tasks.
- The respective datasets used in the studies are discussed to assess the divergence between the different domains.
- Discussions of emerging areas and potential research directions are provided.

ARTICLE INFO

Communicated by K. Liu

Keywords:

Semi-supervised learning
Unsupervised domain adaptation
Medical image segmentation
Deep learning
Frequency

ABSTRACT

Obtaining annotated data for medical image segmentation is a labor-intensive and costly task, due to the need for per-pixel labeling. Semi-supervised learning (SSL) offers a promising solution by utilizing a small amount of labeled data and large-scale unlabeled data. However, most SSL techniques assume labeled and unlabeled data share the same distribution, which is often not the case in real-world settings where domain shifts between datasets are common. Additionally, noisy labels, arising from annotator errors, pose significant challenges, particularly in medical image analysis. While SSL, unsupervised domain adaptation (UDA), and noisy label handling are crucial to clinical applications, current methods typically address these issues separately. In this work, we propose a unified framework that simultaneously tackles SSL, UDA, and noisy label learning. Our approach introduces two key modules: Frequency-Aware Feature Transformation (FAFT) and Adaptive Memory Cross-Attention (AMCA). FAFT transforms encoder features into the frequency domain, separating them into low- and high-frequency (HF) components. We introduce controlled noise to HF components, which capture object edges, and exchange them between labeled and unlabeled data, helping the model learn robust high-frequency representations less sensitive to noisy annotations. To enhance SSL and UDA in a unified manner, we utilize AMCA, which intervenes in the training process of labeled data by incorporating unlabeled data into the learning process. Through extensive experiments, we demonstrate the superior performance of our method in both clean and noisy label scenarios.

1. Introduction

Obtaining annotations for tasks such as segmentation, which require per-pixel labeling [1,2], is both time-consuming and expensive. Semi-supervised learning (SSL) provides an effective solution by leveraging a small amount of annotated data to utilize large-scale unlabeled data [3,4]. Numerous SSL techniques have been introduced for semi-supervised medical image segmentation (SSMIS) [5–7]. However, these approaches typically assume that labeled and unlabeled data share the same distribution, which is often not the case in real-world settings.

Medical images are acquired from different medical centers using various scanning devices, leading to domain shifts that can negatively impact model performance. Domain shifts between labeled and unlabeled data are typically addressed using unsupervised domain adaptation (UDA) techniques [8]. Furthermore, real-world medical images are often affected by various types of noise and artifacts, such as motion blur, electronic noise, and inconsistencies in image acquisition protocols. These factors can degrade image quality, making it challenging to extract accurate diagnostic information [9–11]. Consequently, even if deep learning models are not explicitly trained on degraded images,

* Corresponding author.

Email addresses: suruchi_k@cs.iitr.ac.in (S. Kumari), pravendra.singh@cs.iitr.ac.in (P. Singh).

<https://doi.org/10.1016/j.neucom.2026.134785>

Received 27 April 2026; Received in revised form 3 August 2026; Accepted 9 August 2026

Available online 11 August 2026

0925-2312/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

they should be robust enough to handle such imperfections. In addition, label noise is inevitable in real-world applications and can arise from systematic errors or annotator biases. Since noisy labels degrade the generalization performance of deep neural networks, learning with noisy labels has become a crucial task in medical image analysis [12].

Existing approaches address the above problems separately; however, developing a unified framework that integrates SSL and UDA is a logical next step, as both rely on learning from a mix of labeled and unlabeled data. Moreover, real-world applications require models that can effectively handle label noise, as human annotation errors are inevitable, and managing degraded images during inference is equally crucial. Developing an architecture capable of excelling across all these scenarios remains a challenge. Recently, Wang et al. [13] proposed a generic framework for volumetric SSL that jointly addresses SSL and UDA through an aggregation and decoupling approach. However, as shown in Tables 5 and 7, this method struggles to handle noisy annotations and degraded images effectively.

In this work, we set an ambitious goal of addressing SSL, UDA, and noisy labels within a single framework. To achieve this, we propose a unified approach where we introduce Frequency-Aware Feature Transformation (FAFT) and Adaptive Memory Cross-Attention (AMCA). Specifically, FAFT transforms the learned encoder features into the frequency domain, decomposing them into low-frequency (LF) and high-frequency (HF) components. The LF components capture content, while the HF components encode fine-grained details such as edges and boundaries. We refine the LF components and introduce controlled noise to the HF components before exchanging HF information between labeled and unlabeled data. Since label noise predominantly occurs at region boundaries [14], exchanging HF features between labeled and unlabeled data, along with introducing perturbations to the HF components, allows the model to observe variations of the same object with different HF characteristics. This encourages the model to learn more robust high-frequency representations, making it less sensitive to noisy annotations (Section 5.2).

To enhance SSL and UDA in a unified manner, we utilize AMCA, which intervenes in the training process of labeled data by incorporating unlabeled data into the learning process. Specifically, it extracts channels from the unlabeled features and stores them in a memory bank. The labeled features are then compared with the unlabeled ones on a channel-wise basis, and channels with high similarity are emphasized and used to refine the labeled features. The rationale behind using AMCA in SSL and UDA is that labeled data is significantly less abundant than unlabeled data. By reconstructing labeled features using information from unlabeled data, we introduce greater diversity into the labeled feature space, creating a more challenging feature space. For UDA, AMCA helps by aligning feature distributions between the source (labeled) and target (unlabeled) domains. Since domain shifts cause discrepancies in feature representations, AMCA leverages similarity-based reconstruction to integrate target-domain characteristics into the source feature space. By emphasizing the most relevant target-domain channels, AMCA adapts the source domain to the target and improves adaptation.

Our main contributions can be summarized as follows: We introduce a novel unified framework that simultaneously addresses key challenges in medical image segmentation, including limited annotations, domain shifts, and degraded images at inference time, through a Wavelet-based Cross-Component Exchange (WCCE) mechanism. This work extends our conference version [15], and the additional contributions of this extended version are as follows:

We propose an enhanced approach that, in addition to handling SSL, UDA, and degraded images at inference time, also tackles the problem of label noise within a single unified framework. To mitigate the effects of label noise, we introduce the FAFT module, which enhances robustness to noisy annotations, particularly along object boundaries. Furthermore, we present AMCA, which reconstructs labeled features by integrating the most relevant channels from unlabeled features stored

in a semantic memory bank. This design improves feature diversity for SSL and enhances domain alignment for UDA. Through extensive experiments, we demonstrate the superior performance of our method under both clean and noisy label conditions, validating its effectiveness across diverse real-world scenarios.

2. Related work

2.1. Semi-supervised medical image segmentation

In SSMIS, consistency regularization and pseudo-labeling [16,17] methods are commonly applied. In pseudo-labeling approaches, self-training and co-training methods are employed. Co-training methods are particularly effective in generating pseudo-labels as they utilize multiple views or feature sets of the same image to enhance model performance [5,18]. By training different models on distinct aspects or representations of the data, these methods can capture complementary information. Consistency regularization methods have demonstrated strong performance by encouraging high similarity between the predictions of two perturbed networks for the same input image [5,19,20]. Bai et al. [21] propose a bidirectional copy-paste strategy between labeled and unlabeled data and employ a mean teacher framework to learn from unlabeled samples. VCLIPSeg [22] proposes the incorporation of voxel-wise prompt modules specifically tailored for processing 3D data. These methods have achieved significant success, but they struggle to effectively address the domain gap between labeled and unlabeled data, handle degraded images at test time, or manage noisy labels.

Recent studies have increasingly explored the integration of textual information to enhance semi-supervised medical image segmentation [23,24]. Additionally, several approaches employ multiple decoders to learn more robust feature representations [25–27]. Segment Together [28] proposes a unified semi-supervised framework that jointly learns from multiple datasets and segmentation tasks using task prompts and consistency regularization.

2.2. Unsupervised domain adaptation

Unsupervised Domain Adaptation is similar to semi-supervised learning in that both methods leverage unlabeled data to improve model performance, but UDA specifically addresses the additional challenge of a domain gap between the labeled source domain and the unlabeled target domain [29,30]. Existing methods for UDA use GANs for both feature alignment and image translation between the source and target domains [31,32]. SSL techniques, particularly pseudo-labeling, are also employed in UDA by assigning pseudo-labels to unlabeled data using models trained on labeled source domain data. However, the accuracy of these pseudo-labels can be compromised by the domain gap. Therefore, a key aspect of this approach is to employ various strategies to refine and clean the pseudo-labels [10,33–35]. UDA methods in medical image segmentation have demonstrated promising results, but many rely heavily on GANs. GANs can suffer from increased complexity, primarily due to adversarial training [36]. Additionally, approaches tailored for UDA tend to be less effective in SSL settings [13].

2.3. Noise in medical images

Deep neural networks can produce misclassifications and errors when exposed to noise [9]. In the real world, medical images are often subject to various types of noise and artifacts that occur naturally during image acquisition, transmission, or storage. There has been extensive research dedicated to denoising medical images using both traditional and deep learning methods [37]. However, denoising images affects their spatial and temporal characteristics, often leading to a loss of significant details and altering image contrast [38]. Consequently, there is an urgent need for deep learning models that can perform effectively on both clean and noisy images, even if they are not explicitly trained on these variations.

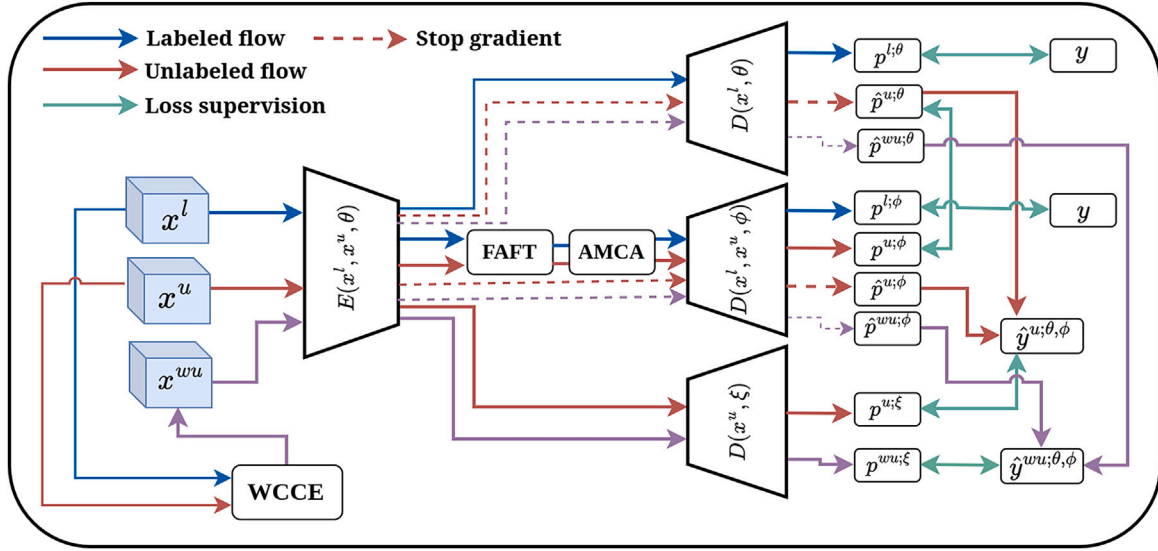


Fig. 1. Illustration of our framework: we utilize one VNet encoder and three decoders. The first decoder, $D(x^l, \theta)$, is trained on labeled data and provides pseudo-labels for the other two decoders. The second decoder, $D(x^l, x^u, \phi)$, leverages both labeled and unlabeled data and generates effective pseudo-labels for the third decoder, $D(x^u, \xi)$. At inference time, we use only $D(x^u, \xi)$ to generate predictions.

2.4. Label noise

The reliability and consistency of medical image labeling are lower compared to natural image labeling. Medical image datasets often suffer from low-quality annotations and pervasive label noise [12]. Some methods utilize regularization terms to reduce overfitting to noisy labels. One such method is label smoothing (LS) [14,39], which addresses label noise by averaging hard one-hot training labels with uniformly distributed soft labels. Additionally, self-supervised pre-training has shown promise for handling noisy labels [40,41]. Several studies have explored redesigning loss functions to mitigate the effects of noisy labels. However, asymmetric loss functions, including Asymmetric Unhinged Loss (AUL) [42] and Mean Absolute Error (MAE) [43], suffer from underfitting and are difficult to optimize [43,44]. Furthermore, loss re-weighting has been introduced to combat noisy labels by dynamically adjusting sample weights during training [45,46]. This approach assigns lower weights to samples likely to be mislabeled and higher weights to those deemed reliable. Recently, Ye et al. [47] leveraged predictive history consistency from a Prediction Memory Bank to identify reliable labels at the pixel level and update them at the superpixel level. To address coarse boundary label issues, they introduced an Edge Detection Branch that explicitly learns boundaries using an edge detection layer. Chen et al. [48] recast noisy label learning as a semi-supervised problem and proposed a Semi-Supervised Distribution Alignment (SSDA) framework that embeds noise-robust distribution alignment into a unified semi-supervised learning scheme.

3. Methodology

3.1. Preliminaries

In SSL and UDA, we are given labeled data $D_l = \{(x_i^l, y_i)\}_{i=1}^N$ and unlabeled data $D_u = \{x_i^u\}_{i=1}^M$. Here, N represents the total number of labeled images, and M denotes the number of unlabeled images. Each image x_i is of size $\mathbb{R}^{H \times W \times D}$, while the corresponding label y_i is represented in $\mathbb{R}^{H \times W \times D \times C}$, where C is the number of possible classes. In SSMIS, the domain is the same for labeled and unlabeled data, while in UDA, the labeled and unlabeled data come from different domains. Our goal in this work is to design a single deep learning model to address both problems.

3.2. Overall pipeline

Similar to our previous work [15], we employ one encoder and three decoders to learn from both labeled and unlabeled data. However, unlike the diffusion V-Net setup with two vanilla decoders, we use a vanilla V-Net encoder and three vanilla decoders, as shown in Fig. 1. $D(x^l, \theta)$ is trained using labeled data, while $D(x^u, \xi)$ learns from unlabeled data, as shown in Fig. 1. Finally, $D(x^l, x^u, \phi)$ leverages both labeled and unlabeled inputs, incorporating Frequency-Aware Feature Transformation (FAFT) and Adaptive Memory Cross-Attention (AMCA). These modules prevent labeled data from dominating the training and effectively bridge the gap between labeled and unlabeled domains in UDA.

3.3. Frequency-aware feature transformation

The intuition behind FAFT is to separately process low- and high-frequency information. The low-frequency (LF) component captures structural content, while the high-frequency (HF) component encodes fine details. To extract LF and HF features, we employ the *discrete wavelet transform* (DWT) with the Haar wavelet. The orthogonality property of the Haar wavelet leads to strong decorrelation [49], enabling a more effective separation of low- and high-frequency components compared to other frequency analysis methods, such as the Fourier transform. Specifically, labeled and unlabeled images are first passed through the encoder $E(x^l, x^u, \phi)$, producing feature representations z^l and z^u . The DWT then decomposes these features into low- and high-frequency coefficients. For a 3D feature z , this results in eight unique frequency coefficients, as illustrated below:

$$LLL, \{LLH, LHL, LHH, HLL, HLH, HHL, HHH\} = \text{DWT}(z)$$

Here, LLL represents the low-frequency (LF) component, which captures the smoothed, coarse representation of the 3D features, preserving general patterns and structures. Conversely, the other components in the equation correspond to the high-frequency (HF) coefficients, which encode finer details and edge information across different spatial dimensions. For simplicity, we refer to LLL as LF and the set of high-frequency components as HF , as illustrated in Fig. 2.

The labeled z^l and unlabeled features z^u , undergo the DWT function, as given in Eqs. (2) and (3). Consequently, each feature is broken

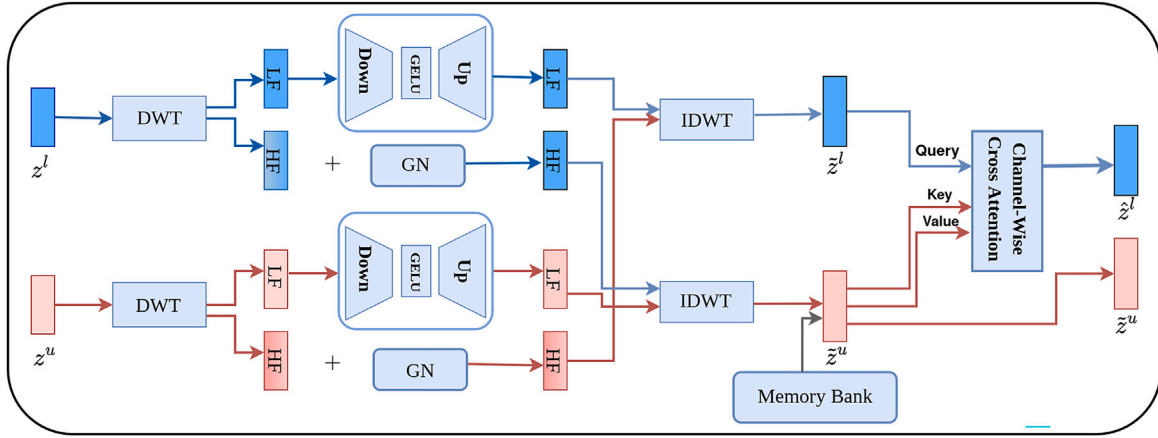


Fig. 2. Overview of our method: we utilize the FAFT and AMCA modules, where FAFT enhances low-frequency features and adds perturbations to high-frequency features. AMCA stores the previous contextual information from the unlabeled data and then uses it to reconstruct the labeled data features through a channel-wise cross-attention mechanism.

down into its LF and HF coefficients. After that we utilize a refiner which comprises a linear down-projection layer, a GELU activation layer, and a linear up-projection layer, as shown in Fig. 2 to refine the low frequency coefficients of labeled and unlabeled data. Furthermore, we introduce controlled noise into the high-frequency components to simulate perturbations that enhance robustness. Specifically, we apply one of the following transformations to the high-frequency coefficients:

$$\tilde{C}_{hf} = \begin{cases} C_{hf} + \epsilon, & \epsilon \sim \mathcal{N}(0, \sigma^2) \\ \alpha C_{hf}, & \alpha > 1 \\ G_{\sigma} * C_{hf}, & \text{(Gaussian blurring)} \\ \min(\max(\beta C_{hf}, 0), 1), & \beta > 1 \end{cases} \quad (1)$$

where $\epsilon \sim \mathcal{N}(0, \sigma^2)$ is random noise, α is sharpening, G_{σ} is Gaussian blurring, and $\beta > 1$ is contrast enhancement. The perturbed coefficients $\tilde{C}_{hf}$, corresponding to the modified high-frequency components, are then used to reconstruct the spatial features via the Inverse Discrete Wavelet Transform (IDWT).

In the reconstruction step, the high-frequency (HF) coefficients of the labeled features are exchanged with those of the unlabeled features, while the low-frequency (LF) coefficients are preserved and vice versa, as illustrated in Eqs. (4) and (5).

$$LF^l, HF^l = \text{DWT}(z^l) \quad (2)$$

$$LF^u, HF^u = \text{DWT}(z^u) \quad (3)$$

$$\tilde{z}^l = \text{IDWT}(LF^l, HF^u) \quad (4)$$

$$\tilde{z}^u = \text{IDWT}(LF^u, HF^l) \quad (5)$$

Label noise predominantly occurs at object boundaries, which are captured by HF components. To address this, we exchange HF coefficients between unlabeled and labeled data and introduce perturbations to the HF components. This allows the model to observe variations of the same object with different HF characteristics, making it less sensitive to noisy annotations.

3.4. Adaptive memory cross-attention

By using FAFT the dominance of biased labeled data is reduced, and the generalization for SSL and UDA, as shown in Table 8. To enhance this further, we propose an adaptive approach that intervenes in the training process of labeled data with unlabeled data. Specifically, we make use of feature channels, which contain richer contextual information. This contextual information, derived from the unlabeled dataset, is subsequently used to reconstruct the labeled data features. This is achieved via

a channel-wise cross-attention mechanism: specifically, the labeled data features serve as the queries, while the unlabeled data features provide the keys and values for this process.

We take the unlabeled features $\tilde{z}^u$, and each feature channel is assigned a class based on its similarity to the pseudo-label. We obtain the pseudo-label from the decoder $D(x_l, \theta)$. These unlabeled features are then stored in a memory bank, organized by class. The size of the memory bank is $\mathbb{R}^{C \times K \times f}$, where C is the number of classes. For each class, K channels are stored, and each channel has the dimensionality of stored features f . To intervene in the flow of labeled data, we apply cross-attention between $\tilde{z}^l$ and the stored unlabeled features $\tilde{z}^u$. We split $\tilde{z}^u$ into the key and value, and then set $\tilde{z}^l$ as the query, as given below:

$$q = \tilde{z}^l W_q, \quad k = \tilde{z}^u W_k, \quad v = \tilde{z}^u W_v \quad (6)$$

where $W_q, W_k, W_v \in \mathbb{R}^{K \times 2K}$ are transformation weights, $\tilde{z}^l, \tilde{z}^u \in \mathbb{R}^{K \times f}$, where f is the flattened feature size and K is the channel dimension. Finally, the channel-wise attention is defined as:

$$\tilde{z}^l = M v^T = \sigma(q^T k) v^T W_{\text{out}} \quad (7)$$

where $\sigma(\cdot)$ denotes the softmax function. Finally, in place of the original $\tilde{z}^l$ features, now $\tilde{z}^l$ is passed to the decoder $D(x^l, x^u, \phi)$ along with $\tilde{z}^u$ to get the final prediction.

3.5. Training

To improve learning stability and robustness, we introduce a small, controllable adjustment to the softmax probabilities. Let $f(x)$ be the model's raw output scores. We first convert these scores into probabilities. Next, we identify the class with the highest probability and increase that value by a constant factor m . After this boost, we re-normalize the entire probability vector by dividing all entries by $1 + m$, ensuring the probabilities still sum to one.

Loss Functions. Our training strategy employs a single VNet encoder and three decoders. We first pass the labeled image through the encoder $E(x^l, x^u, \phi)$ and the decoder $D(x^l, \theta)$, training them using the loss L_{sup}^{θ} . Furthermore, we train the decoder $D(x^l, x^u, \phi)$ on labeled data with the loss L_{sup}^{ϕ} , as given below:

$$L_{\text{sup}}^{\theta} = \frac{1}{B^l} \sum_{i=1}^{B^l} \mathcal{L}_{\text{DiceCE}}(p_i^{\theta}; y_i) \quad (8)$$

$$L_{\text{sup}}^{\phi} = \frac{1}{B^l} \sum_{i=1}^{B^l} \mathcal{L}_{\text{DiceCE}}(p_i^{\phi}; y_i) \quad (9)$$

where $\mathcal{L}_{\text{DiceCE}}(x, y) = \frac{1}{2} (\mathcal{L}_{\text{dice}}(x, y) + \mathcal{L}_{\text{CE}}(x, y))$ represent the conventional cross-entropy and Dice loss, respectively. Furthermore, the decoder $D(x^l, x^u, \phi)$ is trained on unlabeled data, as given below:

$$\mathcal{L}_{\text{unsup}}^{\phi} = \frac{1}{B^u} \sum_{i=1}^{B^u} \mathcal{L}_{\text{DiceCE}}(p_i^{u;\phi}, \hat{y}_i^{u;\theta}) \quad (10)$$

where $\hat{y}^{u;\theta} = \text{argmax}(\hat{p}^{u;\theta})$, and $\hat{p}^{u;\theta}$ represents the prediction of the unlabeled data from the decoder $D(x^u, \theta)$. The argmax function returns the index of the maximum value. Finally, we train the decoder $D(x^u, \xi)$ using only unlabeled data. $D(x^u, \xi)$ utilizes pseudo-labels from both decoders to generate an unbiased and reliable pseudo-label, formulated as follows:

$$\hat{y}^{u;\theta, \phi} = \text{argmax}(\hat{p}^{u;\theta} + \hat{p}^{u;\phi}) \quad (11)$$

After obtaining the pseudo-label, the unsupervised loss for the decoder $D(x^u, \xi)$ is defined as:

$$\mathcal{L}_{\text{unsup}}^{\xi} = \frac{1}{B^u} \sum_{i=1}^{B^u} \mathcal{L}_{\text{DiceCE}}(p_i^{u;\xi}, \hat{y}_i^{u;\theta, \phi}) \quad (12)$$

3.6. Wavelet-based cross-component exchange

After the decomposition of the image into higher and lower frequency components, we utilize the High-Low Frequency Interaction (HLFI) function, which takes the higher frequency components of the labeled image and the lower frequency components of the unlabeled image to form the wavelet-enhanced unlabeled image x^{wu} , as described below:

$$x^{wu} = \text{HLFI}(LLL^u, LLH^l, LHL^l, LHH^l, HLL^l, HLH^l, HHL^l, HHH^l) \quad (13)$$

In the above equation, the superscript u denotes that the frequency components are from the unlabeled image, while the superscript l indicates that the frequency components are from the labeled image. Finally, x^{wu} is converted back to the spatial domain using the inverse discrete wavelet transform (IDWT) (see Fig. 3). To incorporate x^{wu} into our training process, we generate pseudo-labels for x^{wu} in the same manner as

for the original unlabeled images. The final loss is given by:

$$\mathcal{L}_{\text{unsup}_{wu}} = \frac{1}{B^u} \sum_{i=1}^{B^u} \mathcal{L}_{\text{DiceCE}}(p_i^{wu;\xi}, \hat{y}_i^{wu;\theta, \phi}) \quad (14)$$

By utilizing $\mathcal{L}_{\text{unsup}_{wu}}$ in our model training, the model learns to handle images where high-frequency noise patterns are altered, making it less sensitive to noise present in real world medical images.

The final loss for model training is:

$$\mathcal{L}_{\text{final}} = \mathcal{L}_{\text{sup}}^{\theta} + \mathcal{L}_{\text{sup}}^{\phi} + \beta(\mathcal{L}_{\text{unsup}}^{\phi} + \mathcal{L}_{\text{unsup}}^{\xi} + \mathcal{L}_{\text{unsup}_{wu}}^{\xi}) \quad (15)$$

During the testing phase, predictions are obtained using the encoder $E(x^l, x^u; \theta)$ and decoder $D(x^u; \xi)$. The parameter β is set using an epoch-dependent Gaussian ramp-up strategy, as outlined in [50], to progressively enhance the impact of the unsupervised loss.

4. Experimental analysis

Datasets. We performed experiments on two SSMIS datasets, Left Atrium (LA) [51] and Pancreas [52], and one UDA dataset, MMWHS [53], with two settings: MR to CT and CT to MR. The LA dataset contains 3D MRI images, with 80 images for training and 20 images for testing. We adopt the same preprocessing approach as in [13,54]. The Pancreas dataset includes 82 CT images with annotated pancreas regions for segmentation tasks. It consists of 62 images for training and 20 images for testing. The Multi-Modality Whole Heart Segmentation (MMWHS) dataset includes paired CT and MRI 3D images of the same anatomical regions, with 20 images in each modality. Following earlier research [13], we focus on four heart structures: the LVC, AA, LAC and MYO.

Label Noise Settings. To mimic human annotation biases, we follow [47,55], where two parameters, α and β , are introduced to control the extent and intensity of simulated label noise. Specifically, α determines the proportion of training samples whose annotations are intentionally perturbed, reflecting the fact that in real-world scenarios, only a subset of annotations tends to be inconsistent or imprecise. The parameter β , on the other hand, controls the degree of distortion applied to each selected label through morphological or affine transformations, thereby emulating variations introduced by different annotators' subjective boundary judgments. Increasing α results in a larger portion of the dataset being affected by human-like annotation noise, while increasing β amplifies the severity of the distortions in each noisy label. We evaluate our algorithm under noisy conditions with the following settings: $\alpha = 0.5, \beta = 0.5$; $\alpha = 0.7, \beta = 0.7$; and $\alpha = 1.0, \beta = 0.7$.

Implementation Details and Evaluation Metrics. The proposed framework is implemented in PyTorch and runs on a single NVIDIA A5000 GPU. A "poly" decay strategy, as outlined in [64], is employed for learning rate scheduling. We use the same learning rate as in our previous work for all datasets [15]. Additionally, we use a memory size of 50 for AMCA. The Dice coefficient and average surface distance (ASD) serve as standard evaluation metrics for medical image segmentation. Additionally, for the LA datasets, we incorporate the Jaccard index and HD95 metrics, following [33].

4.1. Comparison with SOTA methods for SSL

To thoroughly evaluate our approach in addressing SSL, we compare it with several state-of-the-art (SOTA) methods, as presented in Tables 1 and 2. Specifically, we first compare it with earlier influential methods such as UA-MT [57], SASSNet [58], and DTC [19]. We then include recent SSMIS approaches such as BCP [21] and VCLIPSeg [22], and finally benchmark our method against the unified framework GenericSSL [13]. Our approach consistently performs better

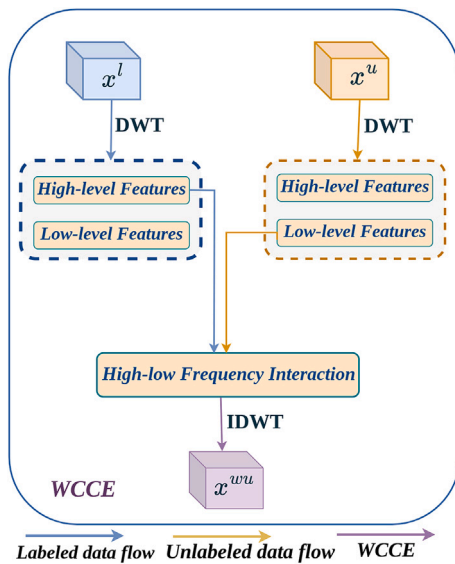


Fig. 3. Wavelet-Based Cross-Component Exchange (WCCE) involves swapping high-frequency wavelet components between labeled and unlabeled images to enhance the model's robustness.

Table 1

Comparisons with SOTA semi-supervised segmentation methods on LA dataset with 5% and 10% labeled ratios.

Method	Scans used		Metrics			
	Labeled	Unlabeled	Dice↑	Jaccard↑	95HD↓	ASD↓
V-Net [56]	4(5%)	0	52.55	39.60	47.05	9.87
V-Net [56]	8(10%)	0	82.74	71.72	13.35	3.26
UA-MT [57]	4(5%)	76(95%)	82.26	70.98	13.71	3.82
SASSNet [58]			81.60	69.63	16.16	3.58
DTC [19]			81.25	69.33	14.90	3.99
URPC [59]			82.48	71.35	14.65	3.65
SS-Net [60]			86.33	76.15	9.97	2.31
BCP [21]			88.02	78.72	7.90	2.15
VCLIPSeg [22]			88.51	79.49	10.48	2.28
SGRS-Net [16]			89.70	81.40	6.68	1.75
GenericSSL [13]			89.93	81.82	5.25	1.86
UDR-3D [15]			90.12	82.03	5.13	1.81
Ours			90.81	82.91	4.58	1.64
UA-MT [57]	8(10%)	72(90%)	87.79	78.39	8.68	2.12
SASSNet [58]			87.54	78.05	9.84	2.59
DTC [19]			87.51	78.17	8.23	2.36
URPC [59]			86.92	77.03	11.13	2.28
SS-Net [60]			88.55	79.62	7.49	1.90
BCP [21]			89.62	81.31	6.81	1.76
MLRPL [61]			89.86	81.68	6.91	1.85
VCLIPSeg [22]			90.59	82.87	6.22	1.61
GenericSSL [13]			90.31	82.40	5.55	1.64
SGRS-Net [16]			90.76	83.13	6.08	1.87
UDR-3D (ours) [15]			91.07	83.67	4.96	1.65
Ours			91.72	84.35	3.74	1.44

Table 2

Comparison of the proposed approach with other approaches on pancreas dataset using 20% labeled data.

Method	Scans used		Metrics			
	Labeled	Unlabeled	Dice↑	Jaccard↑	95HD↓	ASD↓
VNet	12(20%)	0	71.52	57.68	18.12	5.41
UA-MT [57]	12(20%)	50(80%)	76.10	62.62	10.84	2.43
SS-Net [60]			76.20	63.00	10.65	2.68
DTC [19]			76.27	62.82	8.70	2.20
SASSNet [58]			76.39	63.17	11.06	1.42
URPC [59]			80.02	67.30	8.51	1.98
MC-Net + [54]			79.37	66.83	8.52	1.72
SC-SSL [62]			80.76	68.17	6.79	1.73
MCCausSSL [63]			80.92	68.26	8.11	1.53
GenericSSL [13]			79.41	66.20	8.62	1.98
VCLIPSeg [22]			80.56	68.13	6.75	1.49
MLRPL [61]			81.53	69.35	6.81	1.33
UDR-3D (ours) [15]			82.81	71.19	5.77	1.3
Ours			83.43	71.84	5.17	1.2

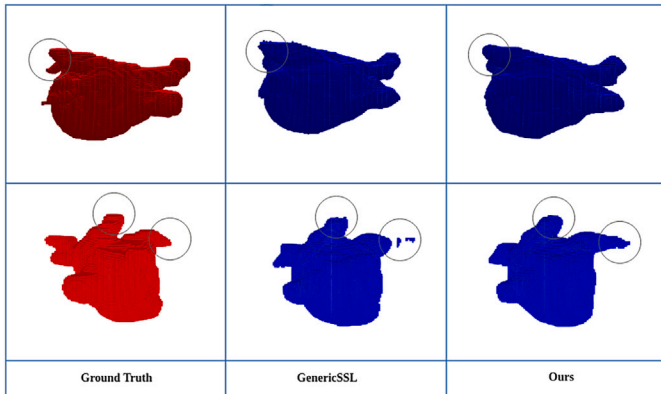


Fig. 4. Visualization results of the SOTA method GenericSSL compared with our method on the LA dataset, utilizing 10% of the labeled data.

Table 3

Comparison of the proposed method with other approaches on the MMWHS dataset for the MR to CT setting.

Method	Dice					ASD
	AA	LAC	LVC	MYO	Avg	Avg
PnP-AdaNet [65]	74.0	68.9	61.9	50.8	63.9	12.8
AdaOutput [66]	65.2	76.6	54.4	43.6	59.9	9.6
CycleGAN [67]	73.8	75.7	52.3	28.7	57.6	10.8
CyCADA [68]	72.9	77.0	62.4	45.3	64.4	9.4
SIFA [32]	81.3	79.5	73.8	61.6	74.1	7.0
DSFN [69]	84.7	76.9	79.1	62.4	75.8	N/A
DSAN [70]	79.9	84.8	82.8	66.5	78.5	5.9
LMISA-3D [71]	84.5	82.8	88.6	70.1	81.5	2.3
ISGAN [72]	89.3	90.2	85.6	71.7	84.2	2.3
GenericSSL [13]	93.2	89.5	91.7	86.2	90.1	1.7
UDR-3D (ours) [15]	85.4	92.9	91.0	95.1	91.1	1.6
Ours	89.2	93.5	92.2	96.0	92.7	1.1

Table 4

Comparison of the proposed method with other approaches on the MMWHS dataset for the CT to MR setting.

Method	Dice					ASD
	AA	LAC	LVC	MYO	Avg	Avg
Supervised Training	82.8	80.5	92.4	78.8	83.6	2.9
PnP-AdaNet [65]	43.7	68.9	61.9	50.8	63.9	8.9
AdaOutput [66]	60.8	39.8	71.5	35.5	51.9	5.7
CycleGAN [67]	64.3	30.7	65.0	43.0	50.7	6.6
CyCADA [68]	60.5	44.0	77.6	47.9	57.5	7.9
SIFA [32]	65.3	62.3	78.9	47.3	63.4	5.7
DSAN [70]	71.3	66.2	76.2	52.1	66.5	5.4
LMISA-3D [71]	60.7	72.4	86.2	64.1	70.8	3.6
SS-Net [60]	62.1	58.4	68.9	51.4	60.2	5.9
BCP [21]	63.6	63.7	70.9	58.0	64.1	4.5
ISGAN [72]	72.9	70.8	80.6	61.3	71.4	3.9
GenericSSL [13]	62.8	87.4	61.3	74.1	71.4	7.9
UDR-3D (ours) [15]	76.2	84.9	91.0	81.6	83.42	2.2
Ours	76.1	85.5	88.3	84.7	83.65	2.0

across both datasets. Additionally, for the LA dataset, we provide qualitative visualization results compared with the SOTA method GenericSSL. As shown in Fig. 4, our method performs better along boundary regions and produces more accurate and visually coherent segmentation results.

4.2. Comparison with SOTA methods for UDA

To thoroughly evaluate our approach in tackling UDA, we compare it with several representative methods, including CyCADA [68] and SIFA [32], as well as recent approaches such as ISGAN [72], and the unified framework GenericSSL [13]. As shown in Tables 3 and 4, our method outperforms existing approaches across both translation directions. Specifically, in the MR-to-CT setting, it achieves a 2.6% improvement in Dice score, while in the CT-to-MR setting, it delivers a substantial gain in segmentation accuracy. Fig. 5 presents qualitative comparisons of our method against several baselines on the MMWHS dataset under both CT-to-MR and MR-to-CT translation settings. As illustrated, our method produces more accurate and visually coherent segmentations.

4.3. Experiments on noisy annotation scenarios

There are no existing methods that jointly handle SSL, UDA, and noisy annotations. Therefore, comparing our approach with methods designed independently for these tasks would be unfair. To ensure a fair comparison, we evaluate our method against GenericSSL, the SOTA method for handling SSL and UDA together. However, since GenericSSL is not explicitly designed to manage label noise, we extend it by integrating two plug-and-play techniques to address noisy annotations. The

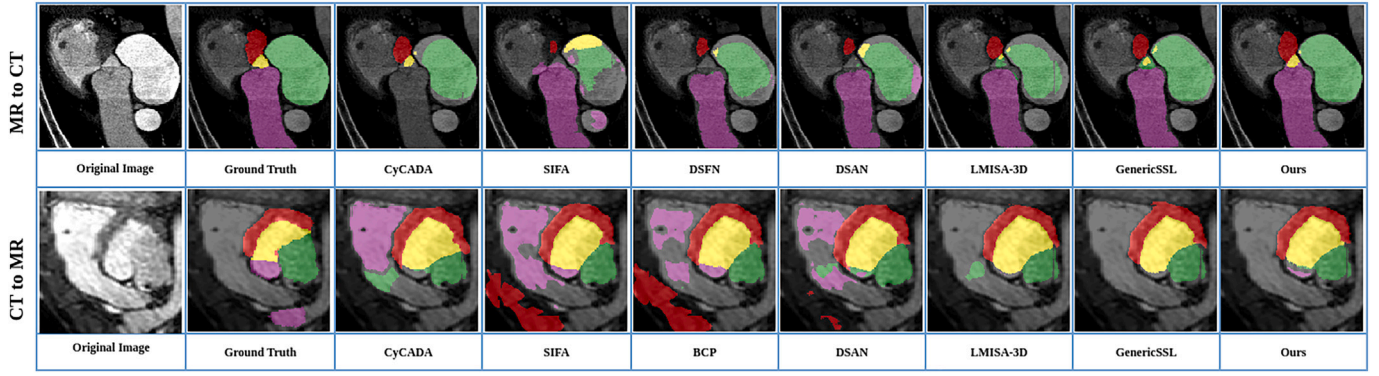


Fig. 5. Visualization results of the SOTA methods with our method on the MMWHS dataset in MR to CT and CT to MR scenarios.

Table 5

Comparative experiments on domain-adaptation datasets when training with noisy labels.

Dataset	MR to CT						CT to MR					
	$\alpha = 0.5, \beta = 0.5$		$\alpha = 0.7, \beta = 0.7$		$\alpha = 1.0, \beta = 0.7$		$\alpha = 0.5, \beta = 0.5$		$\alpha = 0.7, \beta = 0.7$		$\alpha = 1.0, \beta = 0.7$	
	Dice (%) $\uparrow$	ASD (%) $\downarrow$	Dice (%) $\uparrow$	ASD (%) $\downarrow$	Dice (%) $\uparrow$	ASD (%) $\downarrow$	Dice (%) $\uparrow$	ASD (%) $\downarrow$	Dice (%) $\uparrow$	ASD (%) $\downarrow$	Dice (%) $\uparrow$	ASD (%) $\downarrow$
GenericSSL	85.56	4.08	80.83	4.63	78.18	6.14	58.66	13.67	51.52	14.01	45.67	15.87
GenericSSL + 1	86.71	3.56	83.12	3.52	81.38	5.34	62.32	9.34	54.72	12.01	49.78	13.45
GenericSSL + 2	87.7	2.11	84.87	4.84	82.19	5.79	67.38	8.61	62.42	10.54	55.94	12.64
URD-3D (ours)	85.72	5.32	81.52	4.82	79.01	6.47	61.32	12.11	53.12	13.02	48.72	14.25
Ours	90.24	1.92	88.81	2.23	87.20	3.51	79.65	4.22	73.35	6.79	69.73	9.25

Table 6

Comparative experiments on LA dataset when training with noisy labels.

Dataset	LA					
	$\alpha = 0.5, \beta = 0.5$		$\alpha = 0.7, \beta = 0.7$		$\alpha = 1.0, \beta = 0.7$	
	Dice (%) $\uparrow$	ASD (%) $\downarrow$	Dice (%) $\uparrow$	ASD (%) $\downarrow$	Dice (%) $\uparrow$	ASD (%) $\downarrow$
GenericSSL	86.72	3.09	79.76	5.34	77.87	6.12
GenericSSL + 1	87.43	2.71	82.34	3.82	80.12	5.21
GenericSSL + 2	88.76	2.67	85.23	3.76	83.13	4.94
URD-3D (ours)	87.12	3.11	80.24	5.21	77.92	6.37
Ours	90.0	1.72	88.72	2.45	86.43	3.01

first is label smoothing, where hard one-hot labels are averaged with uniformly distributed soft labels, following [14,39]. We refer to this method as GenericSSL + 1. Second, we adopt the approach proposed in [44], which introduces noise to the softmax predictions to enhance robustness against label noise. We refer to this method as GenericSSL + 2. Finally, we compare these approaches with our own [15]. As shown in the Tables 5 and 6, our method outperforms the SOTA method, and even with noise-handling techniques, our approach still achieves superior performance.

To illustrate qualitative results under noisy labels, we compare our method with GenericSSL + 1 on the MMWHS MR→CT setting (Fig. 6). Our method shows stronger robustness to label noise, yielding cleaner boundaries and better structural consistency. Although GenericSSL + 1 improves upon GenericSSL, it still exhibits over-segmentation and misses fine anatomical details.

4.4. Comparison with SOTA method on degraded images

To evaluate the practical robustness of our method, we apply various degradation techniques commonly encountered in medical imaging, including Gaussian noise, motion blur, Rician noise, and Poisson noise,

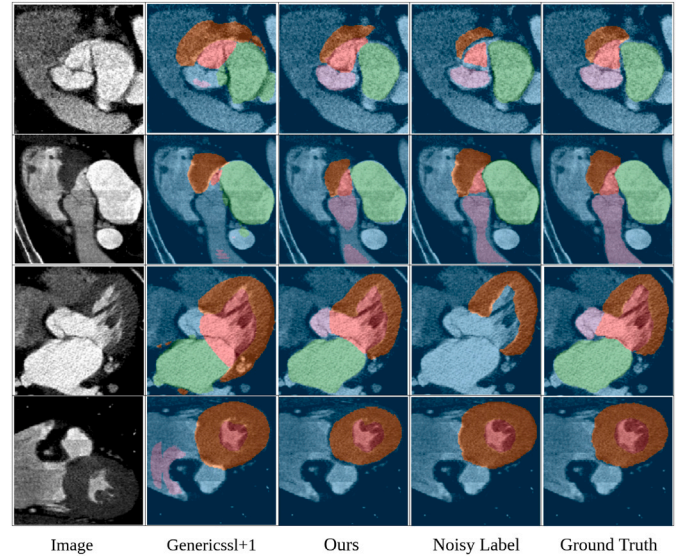


Fig. 6. Visualization results comparing the GenericSSL + 1 method with our approach on the MMWHS dataset (MR-to-CT setting). The “Noisy label” refers to the type of supervision used during training, which contains imprecise annotations.

following the setup used in [15]. We then compare our approach with unified frameworks such as GenericSSL [13] and URD-3D [15] under these degraded conditions. As shown in Table 7, since GenericSSL is not specifically designed to handle degraded inputs, we report its performance both without preprocessing (wo/p) and with preprocessing (w/p) applied to enhance image quality. The preprocessing technique is identical to that used in [15]. As shown in Table 7, our method achieves results comparable to URD-3D [15].

Table 7

Comparison of our method with SOTA methods on degraded images using Dice ($\uparrow$) score, evaluated on both pre-processed (w/p) and non-preprocessed (wo/p) images. Poisson noise is not applicable to MR images.

Dataset	Method	Original image	Gaussian noise		Motion blur		Rician noise		Poisson noise
			$\sigma = 0.1$	$\sigma = 0.2$	$k = 5$	$k = 10$	$\sigma = 0.05$	$\sigma = 0.1$	Scale = 60
MR to CT	Genericssl (wo/p)	90.1	89.72	88.57	89.01	86.67	87.72	85.1	87.41
	Genericssl (w/p)	90.1	89.81	89.01	89.52	88.16	89.26	88.29	88.4
	URD-3D (ours)	91.1	91.02	90.82	90.83	89.51	90.92	90.2	90.31
	Ours	92.7	92.65	92.41	92.56	92.1	91.7	91.6	91.8
CT to MR	Genericssl (wo/p)	71.4	70.55	68.76	68.51	64.85	70.34	65.65	N/A
	Genericssl (w/p)	71.4	70.69	69.13	69.4	67.27	70.91	68.43	N/A
	URD-3D (ours)	83.4	83.37	83.06	82.92	81.74	83.33	82.31	N/A
	Ours	83.6	83.54	83.45	83.01	81.91	83.52	82.62	N/A
LA	Genericssl (wo/p)	90.31	89.81	88.27	89.53	86.0	89.75	87.12	N/A
	Genericssl (w/p)	90.10	89.87	88.61	89.7	86.8	89.86	87.92	N/A
	URD-3D (ours)	91.15	91.01	90.7	90.85	88.21	91.12	91.0	N/A
	Ours	91.72	91.70	91.48	91.45	88.63	91.61	91.21	N/A

Table 8

Ablation study of different components used in our approach on the MMWHS dataset in the CT to MR setting.

VNet-I	VNet-II	VNet-III	WCCE	FAFT	AMCA	Dice $\uparrow$	ASD $\downarrow$
✓						60.1	15.8
✓	✓					64.4	9.3
✓	✓	✓				73.3	7.1
✓	✓	✓	✓			76.6	4.1
✓	✓	✓	✓	✓		78.8	3.7
✓	✓	✓	✓	✓	✓	83.6	2.1

5. Ablation study

5.1. Contribution of different components

Table 8 presents the ablation study, illustrating the impact of all proposed components on overall performance. We first applied self-training to the VNet-I architecture (baseline model), which includes $E(x^l, x^u, \theta)$ and $D(x^l, x^u, \theta)$, achieving a Dice score of 60.1%. To enhance performance, we introduced an additional vanilla VNet decoder, $D(x^l, \phi)$, trained exclusively on labeled data. The pseudo-labels generated by this decoder were then used to guide the training of unlabeled data, forming what we call the VNet-II approach. This modification significantly improved the Dice score to 64.4%. Further enhancement was achieved by adding a second vanilla decoder, $D(x^u, \xi)$, which was trained on unlabeled data using pseudo-labels provided by the other two decoders. We call this framework VNet-III. This led to a Dice score of 73.3%. Incorporating the WCCE module within $D(x^u, \xi)$ further improved the Dice score by 3.3%, reaching 76.6%. Integrating FAFT within the $D(x^l, x^u, \phi)$ decoder further improved the Dice score by 2.2%, raising it to 78.8%. Finally, incorporating the AMCA module provided a substantial boost, culminating in a final Dice score of 83.6% and an ASD of 2.1. This progression clearly demonstrates the cumulative benefits of each component in our approach.

5.2. Contribution of FAFT for noisy labels

Table 9 highlights the contribution of the FAFT module in combating label noise. Without FAFT, the model struggles significantly with label noise, leading to a noticeable drop in performance. However, with FAFT, the model achieves results much closer to those obtained when trained with clean labels. This demonstrates the effectiveness of FAFT in enhancing robustness against noisy annotations. Notably, the improvement in the ASD metric further supports this claim. Since ASD measures the average distance between the predicted and ground-truth boundaries, it serves as a sensitive indicator of boundary precision. Lower ASD values indicate more accurate boundary localization. The substantial

Table 9

Comparison of performance on noisy labels with or without (w/o) frequency aware feature transform (FAFT).

Method	LA		MR to CT	
	Dice (%) $\uparrow$	ASD (voxel) $\downarrow$	Dice (%) $\uparrow$	ASD (voxel) $\downarrow$
w/o FAFT	86.7	3.6	88.2	3.2
with FAFT	90.0	1.7	90.2	1.9

Table 10

Comparison of model parameters and inference time between the SOTA methods and our proposed method in the MR-to-CT setting.

Method	Param. (M)	Inference time (s/iter)	Dice (%) $\uparrow$	ASD (voxel) $\downarrow$
GenericSSL	57.8	3.69	90.1	1.7
URD-3D (ours)	57.9	3.68	91.1	1.6
Ours	67.2	1.68	92.7	1.1

reduction in ASD with FAFT shows that the module effectively preserves fine structural details and mitigates boundary distortions caused by noisy annotations, leading to more reliable and anatomically consistent segmentation results.

5.3. Parameter and time complexity

In this section, we compare the model parameters and time complexity of our method with GenericSSL [13] and our previous method UDR-3D [15]. Our method is based on the VNet architecture and incorporates two additional VNet decoders along with the FAFT and AMCA modules. This results in an increase of only 9.34 M parameters. However, the inference time remains significantly lower compared to these methods. This is because both GenericSSL and UDR-3D employ a diffusion-based encoder-decoder architecture, which is computationally intensive. In contrast, although our framework uses three decoders during training, only a single encoder and a single decoder $D(x^u, \xi)$ are used during inference, leading to substantially faster inference, as shown in Table 10. This makes our model highly suitable for real-world applications.

6. Conclusion

In this work, we address the challenges of SSL, UDA, degraded images at inference time, and noisy label handling in medical image segmentation in a unified manner—problems frequently encountered in real-world clinical applications. We achieve this by leveraging the Frequency-Aware Feature Transformation (FAFT), Adaptive Memory Cross-Attention (AMCA), and Wavelet-Based Cross-Component

Exchange (WCCE). FAFT transforms encoder features into the frequency domain, separating them into low- and high-frequency components, and exchanges high-frequency information between labeled and unlabeled data, thereby enhancing robustness to noisy annotations. Meanwhile, AMCA improves both SSL and UDA by incorporating unlabeled data into the learning process for labeled data, diversifying the feature space, aligning feature distributions across domains, and strengthening cross-domain adaptation. Experimental results demonstrate that the proposed framework consistently outperforms existing methods under both clean and noisy label conditions, providing a robust and unified solution for medical image segmentation tasks involving label noise and domain shifts.

7. Limitations and future work

Although the proposed method demonstrates strong performance in both semi-supervised learning (SSL) and unsupervised domain adaptation (UDA) scenarios, separate training is required for each task. A promising direction for future research is to develop a unified framework that can be trained once and effectively generalize to both SSL and UDA settings. Furthermore, the performance in the CT-to-MR adaptation scenario is lower than that in the MR-to-CT setting. This performance gap is primarily attributed to the higher anatomical clarity and richer structural information available in MR images. Consequently, when MR images are used as the labeled source domain, the learned representations generalize more effectively to CT images. In contrast, CT images provide comparatively less detailed structural information, leading to relatively lower performance when used as the labeled source domain for adaptation to MR images. Although the proposed method still achieves superior performance compared with existing state-of-the-art methods in both adaptation directions, there remains room for further improvement, particularly in challenging cross-modality adaptation scenarios.

CRediT authorship contribution statement

Suruchi Kumari: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Conceptualization. **Pravendra Singh:** Writing – review & editing, Visualization, Supervision, Software, Resources, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets used in this study are publicly available.

References

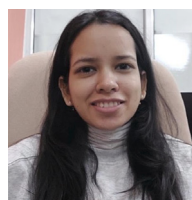
- [1] X. Chen, X. Wang, K. Zhang, K.-M. Fung, T.C. Thai, K. Moore, R.S. Mannel, H. Liu, B. Zheng, Y. Qiu, Recent advances and clinical applications of deep learning in medical image analysis, *Med. Image Anal.* 79 (2022) 102444.
- [2] S. Kumari, P. Singh, Learning with less: a survey of deep learning in medical imaging under varying supervision levels, *Artif. Intell. Med.* (2026) 103375.
- [3] Q. Zeng, Y. Xie, Z. Lu, Y. Xia, Pefat: boosting semi-supervised medical image classification via pseudo-loss estimation and feature adversarial training, in: 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE, 2023, pp. 15671–15680.
- [4] S. Kumari, P. Singh, Integrating orthogonal supervision for sparse semi-supervised 3D medical image segmentation, *Neural Netw.* (2026) 109039.
- [5] R. Jiao, Y. Zhang, L. Ding, B. Xue, J. Zhang, R. Cai, C. Jin, Learning with limited annotations: a survey on deep semi-supervised learning for medical image segmentation, *Comput. Biol. Med.* (2023) 107840.
- [6] S. Kumari, P. Singh, Data efficient deep learning for medical image analysis: a survey, *arXiv preprint arXiv:2310.06557*, 2023.
- [7] X. Ma, S. Xu, J. Zhou, Q. Yang, Y. Yang, K. Yang, S.H. Ong, Point set registration with mixture framework and variational inference, *Pattern Recognit.* 104 (2020) 107345.
- [8] X. Liu, C. Yoo, F. Xing, H. Oh, G. El Fakhri, J.-W. Kang, J. Woo, et al., Deep unsupervised domain adaptation: a review of recent advances and perspectives, *APSIPA Trans. Signal Inf. Process.* 11 (1) (2022).
- [9] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, R. Fergus, Intriguing properties of neural networks, *arXiv preprint arXiv:1312.6199*, 2013.
- [10] X. Ma, Y. Pan, Q. Zeng, M. Lu, B.M. Yezhanuly, B. Matkerim, Y. Xia, Deformable medical image registration with effective anatomical structure representation and divide-and-conquer network, *IEEE J. Biomed. Health Inform.* 30 (2025) 5282–5295.
- [11] S. Kumari, P. Singh, Annotation ambiguity aware semi-supervised medical image segmentation, in: 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE, 2025, pp. 10404–10413.
- [12] J. Shi, K. Zhang, C. Guo, Y. Yang, Y. Xu, J. Wu, A survey of label-noise deep learning for medical image analysis, *Med. Image Anal.* 95 (2024) 103166.
- [13] H. Wang, X. Li, Towards generic semi-supervised framework for volumetric medical image segmentation, in: *NeurIPS*, vol. 36, 2024.
- [14] M. Lukasik, S. Bhojanapalli, A. Menon, S. Kumar, Does label smoothing mitigate label noise? in: *ICML, PMLR*, 2020, pp. 6448–6458.
- [15] S. Kumari, P. Singh, A unified degradation-robust approach to SSL and UDA for 3D medical images, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, 2025, pp. 4428–4436.
- [16] T. Wang, X. Zhang, Y. Chen, Y. Zhou, L. Zhao, T. Tan, T. Tong, Synergy-guided regional supervision of pseudo labels for semi-supervised medical image segmentation, in: *MICCAI, Springer*, 2025, pp. 530–540.
- [17] X. Ma, H. Cui, S. Li, Y. Yang, Y. Xia, Deformable medical image registration with global-local transformation network and region similarity constraint, *Comput. Med. Imaging Graph.* 108 (2023) 102263.
- [18] X. Chen, Y. Yuan, G. Zeng, J. Wang, Semi-supervised semantic segmentation with cross pseudo supervision, in: *CVPR*, 2021, pp. 2613–2622.
- [19] X. Luo, J. Chen, T. Song, G. Wang, Semi-supervised medical image segmentation through dual-task consistency, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, 2021, pp. 8801–8809.
- [20] S. Kumari, H. Singh, P. Singh, Improving medical image segmentation with implicit representation and noisy label robustness, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer*, 2025, pp. 268–278.
- [21] Y. Bai, D. Chen, Q. Li, W. Shen, Y. Wang, Bidirectional copy-paste for semi-supervised medical image segmentation, in: *CVPR*, 2023, pp. 11514–11524.
- [22] L. Li, S. Lian, Z. Luo, B. Wang, S. Li, VCLIPSeg: voxel-wise CLIP-enhanced model for semi-supervised medical image segmentation, in: *MICCAI, Springer*, 2024, pp. 692–701.
- [23] Q. Zeng, H. Luo, Z. Lu, Y. Xie, Z. Wang, Y. Zhang, Y. Xia, Harnessing text insights with visual alignment for medical image segmentation, *IEEE Trans. Med. Imaging* 45 (2025) 477–489.
- [24] Q. Zeng, H. Luo, X. Ma, Z. Lu, Y. Hu, Y. Xia, Exploring text-enhanced mixture-of-experts for semi-supervised medical image segmentation with composite data, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer*, 2025, pp. 226–236.
- [25] Q. Zeng, Z. Lu, Y. Xie, Y. Xia, Pick: predict and mask for semi-supervised medical image segmentation, *Int. J. Comput. Vis.* 133 (6) (2025) 3296–3311.
- [26] Q. Zeng, H. Luo, X. Ma, Z. Lu, Y. Hu, M. Lu, Y. Zhang, Y. Xia, Decoupling target semantics via text-anchored visual contrast for semi-supervised medical image segmentation, *IEEE Trans. Image Process.* 35 (2026) 3228–3239.
- [27] Q. Zeng, Y. Xie, Z. Lu, M. Lu, J. Zhang, Y. Xia, Consistency-guided differential decoding for enhancing semi-supervised medical image segmentation, *IEEE Trans. Med. Imaging* 44 (1) (2024) 44–56.
- [28] Q. Zeng, Y. Xie, Z. Lu, M. Lu, Y. Wu, Y. Xia, Segment together: a versatile paradigm for semi-supervised medical image segmentation, *IEEE Trans. Med. Imaging* 44 (2025) 2948–2959.
- [29] H. Guan, M. Liu, Domain adaptation for medical image analysis: a survey, *IEEE Trans. Biomed. Eng.* 69 (3) (2021) 1173–1185.
- [30] S. Kumari, P. Singh, Deep learning for unsupervised domain adaptation in medical imaging: recent advancements and future perspectives, *Comput. Biol. Med.* (2023) 107912.
- [31] Y. Ganin, V. Lempitsky, Unsupervised domain adaptation by backpropagation, in: *ICML, PMLR*, 2015, pp. 1180–1189.
- [32] C. Chen, Q. Dou, H. Chen, J. Qin, P.-A. Heng, Synergistic image and feature adaptation: towards cross-modality domain adaptation for medical image segmentation, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, 2019, pp. 865–872.
- [33] S. Wu, C. Chen, Z. Xiong, X. Chen, X. Sun, Uncertainty-aware label rectification for domain adaptive mitochondria segmentation, in: *MICCAI, Springer*, 2021, pp. 191–200.
- [34] H. Cho, K. Nishimura, K. Watanabe, R. Bise, Effective pseudo-labeling based on heatmap for unsupervised domain adaptation in cell detection, *Med. Image Anal.* 79 (2022) 102436.
- [35] X. Ma, Q. Zeng, Y. Hu, M. Lu, J. Zhou, Y. Xia, A variational Bayesian algorithm for probabilistic affine and non-rigid point cloud registration, *Pattern Recognit.* (2026) 113645.
- [36] A. Peláez-Vegas, P. Mesejo, J. Luengo, A survey on semi-supervised semantic segmentation, *arXiv preprint arXiv:2302.09899*, 2023.
- [37] W. El-Shafai, S.A. El-Nabi, A.M. Ali, E.-S.M. El-Rabaie, F.E. Abd El-Samie, Traditional and deep-learning-based denoising methods for medical images, *Multimed. Tools Appl.* 83 (17) (2024) 52061–52088.
- [38] T. Dhar, N. Dey, S. Borra, R.S. Sherratt, Challenges of deep learning in medical image analysis—improving explainability and trust, *IEEE Trans. Technol. Soc.* 4 (1) (2023) 68–75.

- [39] J. Wei, H. Liu, T. Liu, G. Niu, M. Sugiyama, Y. Liu, To smooth or not? When label smoothing meets noisy labels, arXiv preprint arXiv:2106.04149, 2021.
- [40] E. Zheltonozhskii, C. Baskin, A. Mendelson, A.M. Bronstein, O. Litany, Contrast to divide: self-supervised pre-training for learning with noisy labels, in: WACV, 2022, pp. 1657–1667.
- [41] J. Chen, R. Zhang, T. Yu, R. Sharma, Z. Xu, T. Sun, C. Chen, Label-retrieval-augmented diffusion models for learning from noisy labels, in: NeurIPS, vol. 36, 2023, pp. 66499–66517.
- [42] X. Zhou, X. Liu, J. Jiang, X. Gao, X. Ji, Asymmetric loss functions for learning with noisy labels, in: International Conference on Machine Learning, PMLR, 2021, pp. 12846–12856.
- [43] A. Ghosh, H. Kumar, P.S. Sastry, Robust loss functions under label noise for deep neural networks, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 31, 2017.
- [44] J. Wang, X. Zhou, D. Zhai, J. Jiang, X. Ji, X. Liu, epsilon-softmax: approximating one-hot vectors for mitigating label noise, in: Neural Information Processing Systems, 2024.
- [45] L. Ju, X. Wang, L. Wang, D. Mahapatra, X. Zhao, Q. Zhou, T. Liu, Z. Ge, Improving medical images classification with label noise using dual-uncertainty estimation, TMI 41 (6) (2022) 1533–1546.
- [46] Z. Mirikharaji, Y. Yan, G. Hamarneh, Learning to segment skin lesions from noisy annotations, in: First MICCAI Workshop, DART 2019, MIL3D 2019, Springer, 2019, pp. 207–215.
- [47] S. Ye, Y. Xu, D. Chen, S. Han, J. Liao, Learning a single network for robust medical image segmentation with noisy labels, TMI 43 (2024) 3188–3199.
- [48] B. Chen, Z. Ye, Y. Liu, X. Fang, G. Lu, S. Xie, X. Li, Towards robust semi-supervised distribution alignment against label distribution shift with noisy annotations, IEEE Trans. Multimed. 27 (2025) 6127–6139.
- [49] V.W. Neo, S. Redif, J.G. McWhirter, J. Pestana, I.K. Proudler, S. Weiss, P.A. Naylor, Polynomial eigenvalue decomposition for multichannel broadband signal processing: a mathematical technique offering new insights and solutions, IEEE Signal Process. Mag. 40 (7) (2023) 18–37.
- [50] Y. Lin, H. Yao, Z. Li, G. Zheng, X. Li, Calibrating label distribution for class-imbalanced barely-supervised knee segmentation, in: MICCAI, Springer, 2022, pp. 109–118.
- [51] Z. Xiong, Q. Xia, Z. Hu, N. Huang, C. Bian, Y. Zheng, S. Vesal, N. Ravikumar, A. Maier, X. Yang, et al., A global benchmark of algorithms for segmenting the left atrium from late gadolinium-enhanced cardiac magnetic resonance imaging, Med. Image Anal. 67 (2021) 101832.
- [52] K. Clark, B. Vendt, K. Smith, J. Freymann, J. Kirby, P. Koppel, S. Moore, S. Phillips, D. Maffitt, M. Pringle, et al., The cancer imaging archive (TCIA): maintaining and operating a public information repository, J. Digit. Imaging 26 (2013) 1045–1057.
- [53] X. Zhuang, J. Shen, Multi-scale patch and multi-modality atlases for whole heart segmentation of MRI, Med. Image Anal. 31 (2016) 77–87.
- [54] Y. Wu, Z. Ge, D. Zhang, M. Xu, L. Zhang, Y. Xia, J. Cai, Mutual consistency learning for semi-supervised medical image segmentation, Med. Image Anal. 81 (2022) 102530.
- [55] S. Li, Z. Gao, X. He, Superpixel-guided iterative learning from noisy labels for medical image segmentation, in: MICCAI, Springer, 2021, pp. 525–535.
- [56] F. Milletari, N. Navab, S.-A. Ahmadi, V-net: fully convolutional neural networks for volumetric medical image segmentation, in: 2016 Fourth International Conference on 3D Vision (3DV), IEEE, 2016, pp. 565–571.
- [57] L. Yu, S. Wang, X. Li, C.-W. Fu, P.-A. Heng, Uncertainty-aware self-ensembling model for semi-supervised 3D left atrium segmentation, in: MICCAI, Springer, 2019, pp. 605–613.
- [58] S. Li, C. Zhang, X. He, Shape-aware semi-supervised 3D semantic segmentation for medical images, in: MICCAI, Springer, 2020, pp. 552–561.
- [59] X. Luo, G. Wang, W. Liao, J. Chen, T. Song, Y. Chen, S. Zhang, D.N. Metaxas, S. Zhang, Semi-supervised medical image segmentation via uncertainty rectified pyramid consistency, Med. Image Anal. 80 (2022) 102517.
- [60] Y. Wu, Z. Wu, Q. Wu, Z. Ge, J. Cai, Exploring smoothness and class-separation for semi-supervised medical image segmentation, in: MICCAI, Springer, 2022, pp. 34–43.
- [61] J. Su, Z. Luo, S. Lian, D. Lin, S. Li, Mutual learning with reliable pseudo label for semi-supervised medical image segmentation, Med. Image Anal. (2024) 103111.
- [62] J. Miao, S.-P. Zhou, G.-Q. Zhou, K.-N. Wang, M. Yang, S. Zhou, Y. Chen, SC-SSL: self-correcting collaborative and contrastive co-training model for semi-supervised medical image segmentation, TMI 43 (2023) 1347–1364.
- [63] J. Miao, C. Chen, F. Liu, H. Wei, P.-A. Heng, Causl: causality-inspired semi-supervised learning for medical image segmentation, in: CVPR, 2023, pp. 21426–21437.
- [64] F. Isensee, P.F. Jaeger, S.A. Kohl, J. Petersen, K.H. Maier-Hein, nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation, Nat. Methods 18 (2) (2021) 203–211.
- [65] Q. Dou, C. Ouyang, C. Chen, H. Chen, B. Glocker, X. Zhuang, P.-A. Heng, Pnp-adanet: plug-and-play adversarial domain adaptation network at unpaired cross-modality cardiac segmentation, IEEE Access 7 (2019) 99065–99076.
- [66] Y.-H. Tsai, W.-C. Hung, S. Schuster, K. Sohn, M.-H. Yang, M. Chandraker, Learning to adapt structured output space for semantic segmentation, in: CVPR, 2018, pp. 7472–7481.
- [67] J.-Y. Zhu, T. Park, P. Isola, A.A. Efros, Unpaired image-to-image translation using cycle-consistent adversarial networks, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 2223–2232.
- [68] J. Hoffman, E. Tzeng, T. Park, J.-Y. Zhu, P. Isola, K. Saenko, A. Efros, T. Darrell, Cycada: cycle-consistent adversarial domain adaptation, in: ICML, PMLR, 2018, pp. 1989–1998.
- [69] D. Zou, Q. Zhu, P. Yan, Unsupervised domain adaptation with dual-scheme fusion network for medical image segmentation, in: IJCAI, 2020, pp. 3291–3298.
- [70] X. Han, L. Qi, Q. Yu, Z. Zhou, Y. Zheng, Y. Shi, Y. Gao, Deep symmetric adaptation network for cross-modality medical image segmentation, TMI 41 (1) (2021) 121–132.
- [71] M. Jafari, S. Francis, J.M. Garibaldi, X. Chen, LMISA: a lightweight multi-modality image segmentation network via domain adaptation using gradient magnitude and shape constraint, Med. Image Anal. 81 (2022) 102536.
- [72] J. Li, Y. Zhang, L. Xu, Y. Yao, L. Qi, ISGAN: unsupervised domain adaptation with improved symmetric GAN for cross-modality multi-organ segmentation, IEEE J. Biomed. Health Inform. 29 (2024) 3874–3885.

Author biography



Pravendra Singh received the Ph.D. degree from the Indian Institute of Technology Kanpur, Kanpur, India. He is currently an Assistant Professor with the CSE Department, Indian Institute of Technology Roorkee, Roorkee, India. He has authored or coauthored papers at internationally reputable conferences and journals, including IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IJCV, CVPR, Proc. Eur. Conf. Comput. Vis., NeurIPS, AAAI, IJCAI, IJCV, *Pattern Recognition*, *Neural Networks*, *Knowledge-Based Systems*, *Neurocomputing*, and others. His research interests include deep learning, machine learning, computer vision, and artificial intelligence.



Suruchi Kumari is currently working toward the Ph.D. degree with the Computer Science and Engineering Department, Indian Institute of Technology, Roorkee, Roorkee, India. She holds a Post-Graduate degree in Computer Science and Technology with a specialization from Jawaharlal Nehru University, New Delhi, India. Her research interests include medical imaging, machine learning, and computer vision. Her ongoing doctoral studies focus on contributing to advancements in annotation-efficient deep learning methods for medical images and computer vision.